

0 2 . 1

Describe how mRNA is obtained from pre-mRNA.

[1 mark]

0 2 . 2

Describe how RNA polymerase joins RNA nucleotides.

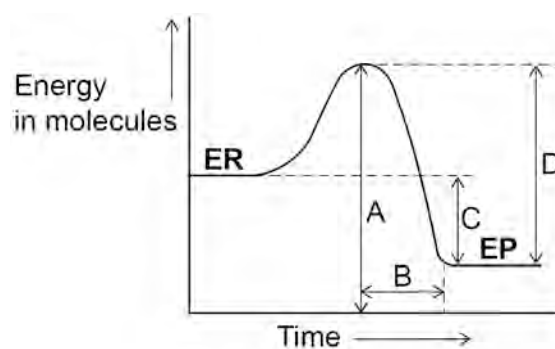
[2 marks]

Question 2 continues on the next page

0 2 . 3

Figure 1 shows the energy contained in molecules during an enzyme-controlled reaction.

Figure 1



Key

ER – Energy in reactants

EP – Energy in products

Which expression represents the activation energy (E_a) of this reaction?

Tick (✓) **one** box.

[1 mark]

A – C

☐

D – C

☐

$\frac{D - C}{B}$

☐

D

☐

0 2 . 4

In 1914, a group of scientists investigated enzyme action.

After studying their results, the scientists reached the following conclusions:

An enzyme is a substance which causes a reaction to occur at “ordinary” temperatures by being present, but **not** by its ability to combine with another substance. An enzyme changes the rate of a reaction without itself being changed by that reaction.

Our current understanding of enzyme action is based on the induced-fit model.

Describe similarities **and** differences between the conclusions from 1914 and our current understanding of enzyme action.

[4 marks]

Similarities _____

Differences _____

Turn over for the next question

Turn over ►

